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3.3.2 Number of research papers in the journals notified on UGC website during the last five years (10)

S.NO	ACADEMIC YEAR	No. of research papers in the journals notified on UGC website during the last five years
1	2020-2021	20
2	2019-2020	25
3	2018-2019	43
4	2017-2018	59
5	2016-2017	32
	Total	179


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Fault Diagnosis of a Double-Row Spherical Roller Bearing for Induction Motor Using Vibration Monitoring Technique

Devarabhotla Sai Chandra  & Y. Seetharama Rao*Journal of Failure Analysis and Prevention* 19, 1144–1152(2019) | [Cite this article](#)127 Accesses | [Metrics](#)

Abstract

Vibration monitoring techniques employing sophisticated instruments are widely used in industries for regular inspection and diagnosis of the health condition of rotating machinery. This paper discusses fault diagnosis of the spherical roller bearing for a 630-kW induction motor in a local industry through vibration monitoring technique using a fast Fourier transform analyzer. The main attraction of this technique is that it can be performed even while the equipment is in normal working condition, thus saving precious downtime and avoiding production loss. Accelerometer sensors are kept on drive and non-drive ends of the motor. Vibration data on motor housing are collected in the triaxial direction in the form of amplitude, spectrum and time waveform signals in full-load and no-load conditions. These signals are analyzed and faults identified from the vibration characteristics. From the data, it is found that the dominating amplitude peaks have appeared in the axial direction at 3960 cpm which is matching with harmonic frequencies of motor RPM. Theoretical calculations were also performed to find the bearing fault frequencies, and it is observed that the vibration readings are matching with sub-harmonics of bearing ball pass frequency of outer race. These investigations and calculations help the maintenance personnel in locating bearing faults and also in detecting the exact cause that leads to bearing deterioration so that a suitable remedy can be suggested, tested and implemented.


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RESEARCH ARTICLE

Synthesis of Benzimidazoles in presence of nanocatalyst Fe_2O_3

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ABSTRACT:

We have an attempt to the synthesis of benzimidazoles by treatment of phenylenediamine with boric acid in water at 120°C in the presence of nanocatalyst, Fe_2O_3 (5 mol%). Initially, we studied the structure and morphology of as prepared nanocatalyst using XRD, FTIR and SEM. Further, the synthesis of benzimidazoles and their derivatives using Fe_2O_3 .

KEYWORDS: Fe_2O_3 , hydrothermal, nanoflakes and benzimidazoles.

INTRODUCTION:

Nowadays, catalysis is a vital utensil of green chemistry because of it offers new synthetic pathways to desired yield by less polluting chemical processes [1, 2]. The superior property of catalyst is ability to be recovered and use in large scale applications at industrial level [3]. The metal oxides homogenous catalysts like rhodium, palladium and iron [4-7] have been reported earlier owing to their possessive properties in view of activity and selectivity; the majority of industrial catalysts remain heterogeneous because of the simplicity of the latter in terms of recovery [3]. The NPs are robust, stable in air, amenable to functionalization, suspendable in many solvents including water or other protic benign solvents; their size, shape and crystallinity can be finely tuned [8].

Due to high activity, selectivity, recyclability and tunability, the metal containing nanoparticles (NPs) are most attractive catalyst. During the past decade, the concept of magnetic NPs (MNPs) has quickly evolved to further simplify the recovery process in catalysis, biology and medicine [9-11].

Based on substituents present at different positions, benzimidazole structure is found in many classes of drugs [12]. Benzimidazole and its derivatives have been widely used in various applications like antimicrobial agents [13], antiviral agent against several viruses such as HIV, influenza and herpes (HSV-1), antitumor, anti-inflammatory, anthelmintic agents and antiprotozoal agents [14-19]. Benzimidazole derivatives are being explored in pharmaceutical industries and substituted benzimidazole derivatives have also been found in the diverse therapeutic applications such as in anti-ulcers, antihypertensive, anti-viral, anti-fungal, anti-cancers and anti-histaminics [20].

Boric acid, a water soluble catalyst and has been found to be effective in various organic transformations such as esterification of hydroxycarboxylic acids [21], aza Michael [22], thia Michael [23], addition, and

Experimental Tensile Strength Characterization Of Metal Matrix Composite Aluminum-6069 Percentage Variation Reinforced With Molybdenum And Coconut Shell Ash

Sasidhar Gurugubelli, Fathimunnisa Begum, Dr.L.S.V. Prasad

Abstract: Aluminium based matrix composites remain the most explored metal matrix material for the development of metal matrix composites. The present work deals with the fabrication of aluminium-6069 alloy metal matrix composites (MMCs) reinforced with molybdenum and coconut shell. The specimens were prepared by taking into consideration of six different weight fractions of aluminium 6069, molybdenum powder and coconut shell ash by stir casting method. The first three specimens were prepared in such a manner that aluminium 6069 and coconut shell ash weight fraction varies by 2% and molybdenum powder weight fraction kept constant by 1%. The next three specimens were prepared in the same manner, coconut shell kept constant by 1%, molybdenum powder and aluminium 6069 varies by 2%. The main purpose of the paper is to study the properties of Aluminium-6069 alloy reinforced with molybdenum and coconut shell ash composites fabricated by stir casting technique and Tensile test was performed by using equipment's such as Universal testing machine. Based on the results obtained, the tensile strength was improved by adding coconut shell ash particles and molybdenum to aluminium-6069 alloy.

Index Terms: Metal Matrix Composites, Aluminium-6069, Molybdenum powder, Coconut Shell Ash, Stir Casting, Mechanical Properties.

1. INTRODUCTION

A composite is a material that consists of constituents produced via a physical combination of pre-existing ingredient materials to obtain a new material with unique properties when compared to the monolithic material properties. This definition distinguishes a composite from other multiphase materials which are produced by bulk processes where one or more phases result from phase transformation. The term matrix and reinforcement are often used. The composites are classified by (a) matrix phase (b) reinforcement.

(a) MATRIX:

The matrix is a percolating "soft" phase in which are embedded the "hard" reinforcements and is completely continuous. In structural applications, the matrix is usually a lighter metal such as aluminium, magnesium or titanium and provides a support for the reinforcement.

REINFORCEMENT:

The reinforcement is embedded into a matrix and it can be continued or discontinued, oriented or disoriented. It does not always serve a purely a structural task but it also used to change physical properties such as wear resistance, friction coefficient or thermal conductivity. It can be worked with standard metal working techniques, such as extrusion, forging, rolling. In addition, they may be machined using conventional techniques, but commonly would need the use of polycrystalline diamond tooling (PCD). The reinforcements used in this present project are molybdenum and coconut shell ash.

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Various works on the application of natural fillers in composites like coconut shell ash, jute, cotton, rice husk wood as the reinforcements in composites have been reported in the literature. P.Rambabu et al. [1] described the alloying and precipitation hardening behaviour, which is the principal strengthening mechanism for Al alloys. Shireesha Y. et al. [2] studied the mechanical properties of composites reinforced by natural fibers and factors which influences the mechanical properties of natural fibre reinforced composites. S.C. Bergsma et al. [3] developed a new 6069 aluminum alloy for application in hot and cold extrusion and forging and observed a favorable formability, with nominal tensile properties after hot or cold extrusion ranging from 394 to 476 MPa UTS, 353 to 448 MPa yield strength, and 14–20% elongation. S.Narayan et al. [4] carried out a systematic tensile and impact testing and hardness measurements on hot forged aluminium metal matrix composites to understand the influence of alloying element and forming process on their mechanical properties. A.Arora et al. [5] were successfully fabricated Metal Matrix Composite (MMC) using an Aluminium plate and Molybdenum powder by Friction Stir Process (FSP) to produce a superficial MMC layer on the Al plate in order to increase the mechanical properties of the as received Al plate. Rajesh A.M. et al. [6] also developed aluminium metal matrix hybrid composite by reinforced Aluminium 7075 alloy with silicon carbide (SiC) and aluminium oxide (alumina) by stir casting method which is less expensive and very effective. B.R. Sunil et al. [7] provided a comprehensive summary of the state-of-the-art in fabricating magnesium-based composites by Friction Stir Processing and also discussed the influence of the secondary phase particles and grain refinement resulted from Friction Stir Processing on the properties of these composites. A.I.O. Zaid et al. [8] investigated the effect of addition of molybdenum either alone or in the presence of titanium to commercially pure aluminium on microstructure and mechanical behaviour after pressing by the ECAP process at room temperature. B.Rebba et al. [9] reported the results of an experimental investigation on the mechanical properties of



Corrosion and cavitation erosion properties of sub-micron WC-Co /Cr₃C₂-NiCr multi-layered coating on aluminium substrates

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chrome
Tungsten carbide-cobalt

Abstract

Marine and automobile components are subjected to corrosion and cavitation erosion during their service. This paper aims to study the use of hard coatings on aluminium substrates with multiple layers of coating comprising of sub-micron sized WC-Co and Cr₃C₂-NiCr for enhancement of hardness, corrosion and cavitation erosion properties. Coatings are applied on aluminium substrates using high-velocity liquid fuel (HVLFF) spray technique and the coating parameters are optimized for best results. The results indicate that multiple layered coating with alternate film coatings of Cr₃C₂-NiCr & submicron WC-Co has significantly improved the hardness of aluminium surface. The resistance to corrosion with multiple layers of coating is found to be exceptional and similar to monolayer Cr₃C₂-NiCr coating. Cavitation erosion tests performed as per ASTM G32 show that coating with multiple layers could resist the erosion of materials under dynamic conditions. The number of layers in the multiple-layer coatings strongly affects the hardness, corrosion and cavitation erosion properties.

1. Introduction

The cast-iron and steel components of IC engines or marine components can be replaced with aluminium parts to reduce their weight significantly. Although aluminium possesses a better heat transfer coefficient, its main drawback is low wear resistance. This predicament limits the scope of using aluminium for the IC engine and marine components. However, by making the aluminium more resistant to wear and by increasing its hardness, it can be used in multiple parts of the automotive engine and marine applications. There are many ways of improving the hardness and wear properties of aluminium by making metallurgical changes such as anodizing, heat treatment, etc. and also by alloying. The other method of improving the properties of aluminium is by using hard coatings that can significantly enhance the hardness thereby improving wear characteristics of the aluminium surface.

Coating with composites of ceramic and metallic materials also called cermet coatings like tungsten carbide with binder material cobalt (WC-Co in 88:12 ratio) and chrome carbide with nickel chrome as a binder (Cr₃C₂-NiCr in 75:25 ratio) had been widely used as thermal spray coatings in automotive and marine industries. Numerous researchers [1-31] investigated the phenomenon of enhancement of hardness and tribological properties of aluminium surfaces coated with diverse types of materials. Materials such as alumina (Al₂O₃), titania (TiO₂), tungsten carbide-cobalt (WC-Co) (88:12), chrome carbide-nickel-chrome (Cr₃C₂-NiCr) (75:25) and the combination of several metallic & ceramic materials have been widely researched as coating constituents.

1.1 Cavitation erosion of marine components

Marine components function under challenging environments due to their encounter with both salt and freshwater, which are corrosive, which is the main reason for the degradation of marine components. The major cause of failure of marine parts is cavitation erosion which is regularly detected in all fluid machinery. Cavitation erosion is an occurrence in which vapor bubbles forming inside a liquid impinge on a metal surface, resulting in the removal of the material. Under cavitation erosion, the substrate is exposed to dynamic fluctuating loads with velocities nearing a few hundred meters per second badly affecting the surface. Hong et al. [10] examined the effect of cavitation erosion on the eroding performance of WC-10% Co-4% Cr coating. They found that the coating was less affected due to corrosive invasion under cavitation erosion compared to substrate materials. Some studies by Krella et al. [14] show that multilayer coatings are more resistant to cavitation erosion compared to monolayer coatings.

Further, as the layers increase beyond a specific limit, there will be deterioration in the cavitation corrosion. This was found to be due to the weakening of the coating bond, owing to a decrease in coating thickness with several coating layers increase. Peat et al. [17] studied three coatings Viz., tungsten carbide, chromium carbide, and aluminium oxide deposited by high-velocity oxy-fuel technique. Investigations made in slurry erosion and dry erosion conditions show that WC-Co is the best choice of coating for resistance to cavitation erosion with chromium carbide failing the test.

Cavity Backed Circular Half Mode SIW Array for Microwave Communications

M. Ravi Kishore, K.C.B.Rao



Abstract: A two element array of circular shaped cavity backed substrate integrated waveguide (SIW) based antenna is proposed in this work. The elements are backed by a dielectric cavity of FR4 epoxy and fed by SIW slot coupling mechanism. Keeping the advantages of the conventional waveguides, the bandwidth of the radiation can be increased by choosing proper dimensions to the slots and circular patches. The two element array configuration in the design contributed to the comfortable uplift of the gain. The impedance matching is achieved by inserting a two arm power divider with pre-calculated dimensions. The accurate formulation of the electromagnetic problem of analyzing the SIW antenna is achieved by using integral equation based methods which can be solved numerically. The designed top layer of the antenna is analyzed with well known Method of Moments (MoM) and the results are compared. The functioning of the antenna is compared in terms of Return losses, radiation pattern and gain. The antenna exhibits 72% of bandwidth with peak gain of 4.2dB in the range of 4.4GHz to 9.9GHz with the resonating frequency of 7.54GHz and well suited for C-band microwave communication applications.

Keywords: Substrate Integrated Waveguide, Circular HMSIW, Cavity backed, Method of Moments, Metallic vias, HFSS Software.

I. INTRODUCTION

Substrate Integrated Waveguide (SIW) technology is the most promising technology for the realization of effective radiating systems for the future microwave communication systems. It is a synthetic rectangular waveguide formed using a dielectric substrate by tightly arraying metalized vias which connect the upper and lower metal layers to form an effective radiating structure. SIW antennas are useful in many applications in the microwave communication systems including C and X-band applications owing to several advantages such as high power handling capacity, low cost, compact profile, high quality factor and integration with MMIC devices. However, the limitation in bandwidth is a never ending issue in the design of microstrip SIW antennas. Many bandwidth enhancement methods have been suggested by the researchers including hybrid mode excitation and exciting the antenna with the slots of dual and multi-resonance characteristics.

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Cavity backed SIW construction is one of the most promising and attractive option in antenna design due to its promising wideband characteristics. This paper proposes a novel two layer cavity backed circular SIW radiator for microwave communication technologies, highlights some of the most important advantages including bandwidth enhancement, gain uplift to the desired levels. The design of the circular array requires different steps including optimization of the radiator dimensions together with the HMSIW cavity, the organization of the coupling between the array elements, and the efficient design of the feed network to energize both the elements. A substrate integrated waveguide (SIW) antenna array consisting two SIW phase shifter sections with two inserted metallic posts operating at 10 GHz is presented in [3]. The phase shift is obtained by changing the positions of two reactive posts inside each element of the array. An ultra wideband and elevated gain cavity-backed square 4X4 array antenna is designed [4] by highlighting the higher-order SIW structure with slot elements in the bottom layer of the antenna. To advance the aperture efficiency, several bowtie and planar slot cavity-backed antennas have been designed [5]-[7]. A Dielectric resonating antenna (DRA) array fed by a cavity-backed SIW is designed and analyzed to radiate in desired directions [8].

A compact monopulse array with double-layer 8-cells SIW antenna has been developed using optimum feed networks with enhanced microstrip to SIW transitions [9]. A phased slot array antenna with wide angle scanning has been demonstrated with decreased gain fluctuation in the microwave operating band [10][11]. Two QMSIW antennas have been proposed for WLAN and WBAN applications [12] and performance of the antennas are successfully tested on a human body.

The design of HMSIW based dual-band antennas are demonstrated in [13 - 15], which has good antenna performance characteristics. Considering all the advantages of Cavity backed radiating structures, the present paper demonstrates the design and analysis of a two element half circular shaped antenna array fed by using slot coupling mechanism. The proposed antenna is simulated by using Full-wave based High Frequency Structure Simulator (HFSS) software.

This paper is prearranged as follows. Section II describes methodology followed for design of cavity backed circular half mode SIW array. Section III describes the fullwave analysis of the upper layer of the proposed radiating structure. Section IV demonstrates the results elucidating antenna radiation characteristics and finally Section V is conclusion.

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Design of Dual Slotted Microstrip Patch Antenna for S- Band Applications

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Abstract

This prospectus presents a slotted microstrip patch antenna with an integrated ring and plus shaped slots. This slotted antenna is designed to operate in S-Band. With the inclusion of slots to the normal patch antenna, performance of the antenna is improved when compared to the antenna without slots. For the above mentioned two slotted shapes, parameters like gain, return loss, VSWR and resonant frequency were depicted and tabulated. RT Duroid with Relative permittivity and loss tangent of 2.2 and 0.0004 respectively was chosen as the dielectric substrate.

Keywords; Ansoft HFSS, Gain, Microstrip Patch Antenna, Return loss, RTDuroid.

I. INTRODUCTION

An MPA is a narrow band, wide beam antenna that can be fabricated by etching the metallic radiating element (patch) to an insulating substrate material. Due to its compactness, minimal cost, extremely light weight and convenient integration with Radio frequency devices, microstrip patch antenna is extensively being used in the field of communication and space applications. The conducting element can take any arbitrary shape but the most preferable one is a patch in rectangular shape because of ease of analysis and fabrication. The amount of radiation from patch antenna can be characterized by dielectric constant. Usually low dielectric values are chosen for maximum radiation. Sophisticated Radar and ASR systems require multiple converging frequencies in order to operate at multiple bands. Hence, this paper is focused on three different frequencies in S- band designation.

Circular polarization is the key factor for satellite communication and space borne applications. Introduction of slots on the patch or fabricating varactor diode or cutting edges can result circular polarization. With reference to the above mentioned

facts, a single band antenna based on circularly polarization and diagonally symmetric strips have been designed in different slot shapes. The triple band antenna is controlled based on Plus and Ring slots. Coaxial feeding is used as a feeding mechanism for the design.

II. MATHEMATICAL BACKGROUND

The proposed antenna was designed with RT Duroid as a dielectric material. As the relative permittivity of the chosen material lies in the lower range of $2 \leq \epsilon_r \leq 12$, it is expected to produce a appreciably good amount of radiation. In order to calculate the dimensions of proposed antenna, transmission line model equations were considered. From the known values of relative permittivity and operating frequency, then the following steps can be performed to obtain the dimensions.

A. Patch Height

The height of the patch can be determined by the formula

$$h = \frac{0.3c}{2\pi f_0 \sqrt{\epsilon_r}}$$

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Deep Learning-Based Segmentation of the Coronary Artery Region in the Cardiac MRI Images Using U-Net Model

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Abstract

The availability of medical imaging technology since the recent few years in the province of computer vision has helped the doctors to visualize and analyze the interior parts of the body with ease. Previously, classical methods have been employed to deal with medical image analysis. Later with the emergence of deep learning concepts and techniques, essentially semantic segmentation, its use has rapidly increased and became one of the most preferred choices of the researchers. That too, in cardiac pathologies, automated segmentation of known object of interest in cardiac images from MRI datasets is a very crucial step for timely diagnosis. So, after the thorough research it is perceived that, U-Net architecture which is based upon deep fully convolutional encoder-decoder model has been accredited for its triumph in the biomedical image processing community. In this work, to meet the requirements of the system objective, the implementation of detection and segmentation of coronary artery on the cardiac images of MRI modality has turned out to be rewarding using U-Net architecture. Evaluation metrics presented in this paper, showcases that this model achieved 96.62% Mean IoU, 77.95% precision, 81.72% recall and 89.22% F1-score on the test dataset.

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I. INTRODUCTION

Over the last few decades there have been efforts to train machines to replicate the human brain which gave rise to the concept of machine learning (ML) [1] where one can train machines to make human like decisions by recognizing certain features. In other words, ML quantifies the contents in the images employing hand-crafted features. Feature extraction descriptors such as texture, shape and color are applied for all the images using some ML algorithm that returns a feature vector (a list of numbers). This resultant feature vectors are then

directed as inputs to the machine learning architectures.

Since 2006, a complex yet effective technique termed deep learning (DL) [2] which is the leading-edge in the manifestation of neural networks (NN), has transpired as headway in the research of ML. It has been impacting a vast scale of works in the real time processing of information. The advantage of DL is that the program automatically builds the feature set to classify data, as opposed to most traditional ML algorithms, which require intense time and effort of data scientists. The DL

An Efficient Fake News Detection System Using Machine Learning

A.Lakshmanarao, Y.Swathi, T. Srinivasa Ravi Kiran

Abstract: Social media plays a major role in several things in our life. Social media helps all of us to find some important news with low price. It also provides easy access in less time. But sometimes social media gives a chance for the fast-spreading of fake news. So there is a possibility that less quality news with false information is spread through the social media. This shows a negative impact on the number of people. Sometimes it may impact society also. So, detection of fake news has vast importance. Machine learning algorithms play a vital role in fake news detection; Especially NLP (Natural Language Processing) algorithms are very useful for detecting the fake news. In this paper, we employed machine learning classifiers SVM, K-Nearest Neighbors, Decision tree, Random forest. By using these classifiers we successfully build a model to detect fake news from the given dataset. Python language was used for experiments.

Index Terms: fake news, machine learning, python.

I. INTRODUCTION

Social media is a very fast-growing thing from the last decade. Most of the information generating today come from social media. In some cases, social media can have the capability of spreading the news more quickly than newspaper Media, TV media. It can cover news that was unable to cover by other media. Generally, this kind of fake news is created to promote some agenda [8]. Fake news also caused issues like sarcastic articles or fabricated news or some cases pretending to plan government propaganda. It is very possible that 2 different articles that are similar in their number of words may be opposite in their meaning. The data science community has responded by taking action against the problem. Kaggle conducted a competition named as the "Fake News Challenge" and Facebook is using Artificial Intelligence to filter fake news stories out of users' feeds. Fake news classification is a text classification problem with a straightforward proposition. There is a challenge to design a model that can detect fake news or real news. It is very possible that 2 different articles that are similar in their number of words may be opposite in their meaning. The data science community has responded by taking action against the problem. Kaggle conducted a competition named the "Fake News Challenge" and Facebook is using Artificial Intelligence to filter fake news stories out of users' feeds.

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II. LITERATURE SURVEY

Misleading articles studied by Conroy, Rubin, and Chen [1]. They have shown that simple content n-grams and shallow parts of speech tagging are not sufficient for classification methods. Feng, Banerjee, and Choi [2] got more accuracy deception classification tasks. Junaed Younus Khan et.al [3] applied online N-gram analysis and machine learning techniques for the detection of fake news. Three types of fake news identified by Rubin et al. They applied various text analysis and predictive analysis methods. News is generally related to text data. So, Text data handling can be done by Machine Learning Natural Language Processing (NLP) techniques. For detecting fake news, ML NLP is a very useful tool. Hadeer Ahmed [7] used Support Vector Machines with Term Frequency-Inverted Document Frequency (TF-IDF) as feature extraction method and got good accuracy. Shlok Gilda [9] et.al applied different machine learning techniques for fake news detection.

A various number of machine learning algorithms are applied for fake news detection [5] earlier. Each algorithm has its advantages. So comparing all algorithms is necessary. Dataset is tested using Support Vector Machines, Stochastic Gradient Descent, Gradient Boosting, Bounded Decision Trees, and Random Forests. We propose a method for "fake news" detection and ways to apply it on Facebook, one of the most popular online social media platforms. This method uses the Naive Bayes classification model to predict whether a post on Facebook will be labeled as REAL or FAKE.

Raw datasets collected for fake news detection usually contain some noise such as missing values. The performance of any machine learning depends on the input data. So data preprocessing step plays a vital role before applying a machine learning algorithm. Machine Learning Data preprocessing handles missing values efficiently. Specifically, we have successfully handled the missing values problem by using data imputation for both categorical and numerical features. Selecting a dataset is an important step as the complete process is dependent on the fields, records, and data of the dataset. The dataset we used is from kaggle fake news challenge -1. The data is derived from the Emergent Dataset created by Craig Silverman. It has three fields namely headline, body text, and label. The label shows whether the news is classified as fake or real. Conroy[6] et.al constructed a fake news detection system with machine learning naïve Bayes classifiers.

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Classification of Biomedical Images Using VGG16 Deep Learning Architecture

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Abstract

In recent years, the technical advances and improvements in the field of biomedical imaging techniques has provoked increasing interest to resolve the previously intractable problems. From the standpoint of medical arena, the task of classifying various biomedical MRI scans under their appropriate categories is quite tedious owing to the distinct characteristics of these images. That too, with the large-scale influx of medical reports, there is a wide necessity for medical data to be automated so that it can be easily organized and accessed in the databases of hospitals and clinics. This mode of execution is conducive and speedy for the primary diagnosis of the patients while saving plentiful of time in categorizing them. Lately, with the evolution of deeper networks, a derivative from the notion of artificial neural networks known as deep learning has emerged. VGG16 model which adopts the perception of deep learning has performed fairly well to fulfill the objective of this paper. In this paper, 3 different datasets of MRI modality of 3 various interior body structures namely knee joint, lumbar spine and cardiac are considered. The motive is to classify these images into 3 separate classes using VGG16 model. This model is applied on the medical dataset comprising of 328 MRI scans and subsequently followed by data augmentation to increase the size of the data. The model is implemented in keras as front-end library with TensorFlow framework. The training on this dataset to generate a custom model file on GTX 1070 video card consumed 2 hours yielding 99.62%, 98.48%, 96.6% training, validation and testing accuracies respectively.

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Keywords; Deep learning, Image classification, artificial neural networks, biomedical image classification, keras, tensorflow and VGG16

I. INTRODUCTION

Motivation

Healthcare sector is of higher priority when compared to other industries and is entirely different from them. Irrespective of price, people seek for highest standard of services and care. In most of the cases, medical experts interpret the medical images and there exists wide variations across various interpreters owing to the complexity and subjectivity of those images.

In the field of computer vision, extracting and classifying the features of the images is a significant and essential step. This is accomplished by traditional machine learning (ML) techniques and other shallower neural network models until few years ago. In recent times,

Deep learning (DL) [1], a subfield of ML makes use of convolutional neural networks (CNNs) [2] which are firmly established as a powerful method for image recognition. The fundamental building block of DL is CNNs which discovers and learns the latent features of the images itself. The baseline


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3.3.2-143

Analysis of Malicious Facebook Applications Using Machine Learning Algorithms

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ABSTRACT: Online social networks (OSN's) are one of the most widely used technology because of third party apps. Most of the people use social media to upload photos and posts. This gave a chance to hackers to use social media applications for spreading spam and malware. So, we come across a large amount of the applications in our dataset are malicious. Malicious applications vary from benign applications in various features. So, in order to analyze we first identify the set of features that help us differentiate malicious applications from benign applications. Suppose for example, malicious applications will frequently have the identical name of benign applications. We consider mainly two parameters called application name similarity and posted link similarity. We use Naïve Bayes, Support Vector Machine (SVM) and Random Forest to classify between application name similarity and posted link similarity and based on these results we will be able to detect the malicious post. Hence, we use this analysis to reduce the use of malicious applications result accuracy. Together with these parameters the projected framework produces enhanced outcome prediction..

KEYWORDS: Malicious, Machine Learning, Support Vector Machines, Random Forest, Naive Bayes algorithms.


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Reviews Classification

B. Aparna, A. Nagarathnam

In this advanced age where the internet is growing quickly, the presence of the internet can make it simpler for the traveler to discover any data. In the hotel tourism sector, the use of the internet is very helpful for the promotion of their business. There is a vast number of hotel reviews available on the internet. Based on these reviews, hotel owners take advantage of improving facilities. With the accessibility of a large number of reviews, it is difficult to see everyone's review to identify whether they contain positive or negative reviews. In this paper, we proposed machine learning techniques for hotel review classification. The dataset is taken from Kaggle, which contains 10000 reviews. We applied ML natural language processing techniques and classification algorithms on the dataset and achieved an accuracy of 94%. Python language was used for implementation of all experiments.

Index Terms: Hotel Reviews, Machine Learning, Python.

I. INTRODUCTION

At the point when the travelers need to pick an agreeable lodging for their excursion, they will search for surveys from different explorers. Online visitor audits are basic to what's to come of their property. Reviews accessible on the internet are increasingly significant, genuine and nitty-gritty than the audits found in inn leaflets. In any case, with a large number of reviews, hotel owners can't comprehend and abridge all the available reviews whether they contain positive sentiments or negative sentiments. Some survey sites just give appraisals that are viewed as not objective and thusly can't be utilized as correlations between reviews. The rating framework valuation is unique concerning the composed surveys so it can't be utilized for lodging examination because the data gave isn't clear. The hotel customers can express their emotions and reviews on online surveys. By playing out the opinion mining and sentiment analysis on these reviews, we can foresee the rating of that hotel. A good recommender framework is required for producing the ratings clearly and exactly. For a lodging business, reviews about different viewpoints like Maintenance, Food, Accommodation, Room

neatness, Response from the staff are major items for the recommender framework. The Customer's feelings with respect to a lodging depend upon the facilities he/she got from that hotel, much the same as tidiness, area of the lodging, administrations gave by the inn like free Wi-Fi, multilingual staff, bar/relax, keeping an eye on, and wheel seat and so on. The assessments can be communicated as fantastic, great, normal, poor, terrible, etc. Generally, customers need to express their feelings with rating values. The hotel surveys are given only by clients who have reserved a spot at a specific lodging. A recommendation system can give recommendations to a solitary thing or a succession of things. A solitary thing (single thing) is largely utilized at web-based business sites to discover the deal and result of one single thing while the succession recommender framework for a succession of things is utilized to anticipate the matter of any association; for example, lodging business, tidiness, administrations, and these qualities are utilized. The recommender framework is considered as a decent one that can distinguish the fitting clients, their inspiration, desires and the objectives. Machine learning doesn't allude to only certain something; it's an umbrella term that can be applied to a wide range of ideas and strategies. Understanding AI implies being acquainted with various types of model investigation, factors, and calculations. Machine Learning models work with mathematical operations. So, the datasets used for machine learning must be numeric. Even though, the dataset contains categorical features they are converted to numerical values before applying ML techniques. But hotel reviews purely contain textual data, not numeric data. For handling textual data, Machine learning provides Natural Language Processing (NLP) techniques. After applying various NLP techniques, finally, textual data is converted to numeric. Then, data is ready for applying ML techniques.

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Hotel Reviews Classification

B. Aparna, A. Nagarathnam

Abstract: In this advanced age where the internet is developing quickly, the presence of the internet can make it simpler for the traveler to discover any data. In the hotel tourism sector, the use of the internet is very helpful for the promotion of their business. There is a vast number of hotel reviews available on the internet. Based on these reviews, hotel owners take advantage of improving facilities. With the accessibility of a large number of reviews, it is difficult to see everyone's review to identify whether they contain positive or negative reviews. In this paper, we proposed machine learning techniques for hotel review classification. The dataset is taken from Kaggle, which contains 10000 reviews. We applied ML natural language processing techniques and classification algorithms on the dataset and achieved an accuracy of 94%. Python language was used for implementation of all experiments.

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Estimation of Compressive Strength of Concrete with Machine Learning Algorithms

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Abstract

One of the sustainable materials used for the construction field is concrete. It is fire resistant and also gives long service life. But high-performance concrete is necessary for achieving the required goals. The design of concrete involves multiple parameters like cement, water, Fly Ash, etc. The selection of suitable ingredients needs special attention. The final strength of the product depends on the proportions of these parameters. Concrete compressive strength prediction involves tedious and it takes a lot of time. We need to use modern methods to overcome these difficulties and provide quality. In India, most of the constructions are based on concrete. In this paper, we proposed machine learning algorithms for the prediction of compressive strength. A dataset from the UCI Machine Learning repository is used for conducting experiments. The dataset contains 1030 specimens with 9 features. We applied different machine learning algorithms and achieved better prediction results.

Keywords: Concrete, UCI repository, Machine Learning

1. Introduction:

Cement is a composite material and it is the most utilized constructing material for its dependable nature, quality, and unwavering quality. Compressive quality is the most significant properties of concrete as it opposes the compressive pressure made by the structure. In solid development, it is important to guarantee the compressive quality,

Diagnosis of Various Thyroid Ailments using Data Mining Classification Techniques

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Abstract- Classification is one of the most considerable supervised learning data mining technique used to classify predefined data sets the classification is mainly used in healthcare sectors for making decisions, diagnosis system and giving better treatment to the patients. In this work, the data set used is taken from one of recognized lab of Kashmir. The entire research work is to be carried out with ANACONDA3-5.2.0 an open source platform under Windows 10 environment. An experimental study is to be carried out using classification techniques such as k nearest neighbors, Support vector machine, Decision tree and Naïve bayes. The Decision Tree obtained highest accuracy of 98.89% over other classification techniques.

Index Terms- Thyroid disease, K-Nearest Neighbor, Support Vector Machine, Decision Tree, Naïve Bayes.

I. INTRODUCTION

Classification techniques play a vital as well as major role in analyzing survivability of diseases and providing facilities to reduce the cost to the patients. Now-a-days, Disease diagnosis has become very crucial because of occurrence of so many diseases every year.

People from all over the world have been suffering from various health issues like diabetes, heart disease, typhoid, tuberculosis, kidney disease etc [16] [17]. Beside these health issues , thyroid disease have also been detected worldwide and thus become a serious endocrine health problem and an issue of concern. It is expected that in India about 42 million people suffer from thyroid disorders [2]. As per recent studies, women are 5 to 8 times more prone to thyroid disorders than men worldwide. It is caused by the improper secretion of thyroid hormones released from the thyroid gland which is one of the important organ located in the front of the neck and below the Adam's apple of our body. The secretion of thyroid hormones from the thyroid gland are of two types i.e. levothyroxine or T4 and triiodothyronine or T3. These hormones help in production of balanced amount of proteins, regulating the temperature of body, and maintaining overall production of energy [18]. Thyroid disease occurs when thyroid gland stop to functioning properly and are mainly divided into hypothyroidism and hyperthyroidism [4].

The excess and deficient secretion amount of thyroid hormone causes hyperthyroidism and hypothyroidism respectively. The common symptoms of hyperthyroidism are sudden weight loss, rapid heartbeat, nervousness, etc. and hypothyroidism has weight gain, tiredness, weakness, feeling cold etc. One of the most common cause occurred due to hyperthyroidism is graves' disease [3]. The underestimated thyroid disease causes thyroid diseases. The Decision Tree outperformed over other techniques.

The rest of the paper is followed as: - Section 2 represents related work in diagnosis of thyroid storm and myxedema which may lead to death [12].

In this research work, a classification model is trained using classification algorithms like K- nearest neighbor (KNN), support vector machine (SVM), decision tree (DT) and Naïve bayes (NB) for the diagnosis of thyroid diseases. Section 3 contains dataset and methods. Section 4 represents the results and discussion. Section 5 contains conclusion and at last references are mentioned.

II. MATERIALS AND METHODS

2.1 Data Collection

The data used for this work was collected from Ibadan Synoptic Airport through the Nigerian Meteorological Agency, Oyo State office. The case data covered the period of 120 months, that is, January 2000 to December 2009. The following procedures were adopted at this stage of the research: Data Cleaning, Data Selection, Data Transformation and Data Mining.

2.2 Data Cleaning

In this stage, a consistent format for the data model was developed which took care of missing data, finding duplicated data, and weeding out of bad data. Finally, the cleaned data were transformed into a format suitable for data mining.

2.3 Data Selection



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Determination of Aniline and Epichlorohydrin as a Genotoxic Impurity in Levodropropizine by Gas Chromatography

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Abstract

A purpose of the gas chromatographic method has been developed for the identification and quantification of Aniline and Epichlorohydrin in Levodropropizine drug substance. The method was validated as per ICH guideline in terms of specificity, LOD, LOQ. Method precision and accuracy. The LOD and LOQ values for Aniline and Epichlorohydrin were found to be 0.83 ppm and 2.5 ppm respectively.

Keywords: Development, Validation, Aniline, Epichlorohydrin, Levodropropizine, genotoxic impurity, and Gas chromatography

I. INTRODUCTION

Aniline and Epichlorohydrin are used as raw materials in the manufacturing of Levodropropizine. Aniline is Colorless liquid with Aromatic amines like odor with density 1.022 g/mL and a boiling point of 184°C, chemical name is Phenylamine, it has an empirical formula C₆H₇N, and it has a molecular weight of 93.13 g/mol. Epichlorohydrin is Clear Colorless liquid with density 1.181 g/mL, and boiling point of 118°C chemical name is Chloromethyloxirane, it has an empirical formula C₃H₅ClO, and it has molecular weight of 92.52 g/mol

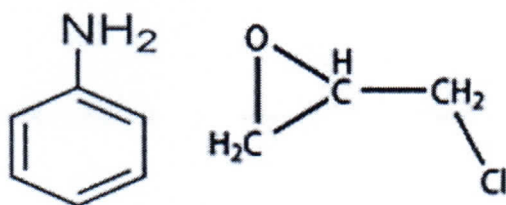


Figure.1 Figure.2

Literature survey reveals that few chromatographic methods HPLC [1-8] and GC-MS [9-10], has been reported for the estimation of Levodropropizine. However, there is no method available for the quantification of Aniline and Epichlorohydrin in Levodropropizine by gas chromatography. In the present work, we have developed a new, simple, precise, accurate method for determination of Aniline and Epichlorohydrin in Levodropropizine by gas chromatography in bulk drug. In synthesis of Levodropropizine, Aniline and Epichlorohydrin are impurities, which is a genotoxic alert. Hence, an analytical method was developed for Aniline and Epichlorohydrin in Levodropropizine, which is a stability indicating method.

II. EXPERIMENTAL

Chemicals

Aniline and Epichlorohydrin were purchased from Aldrich chemicals; GC grade n-Hexane (C₆H₁₂) was purchased from Rankem. Samples of Levodropropizine were received from the local market.

**Interpreting the Confluence of Eco-feminism and Indian Literature in the Literary
Creative of Vandana Shiva**

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Basic Science & Humanities

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Abstract

Since time perennial the world has witnessed a unique connection between the manipulation and dilapidation of the natural world and the subservience and coercion of women. Man through the ages has infuriated being in terms with his corporeal environment as well harness its immense power to solve his problems, without rescinding it. The Hindu scriptures, epics and legends abound with instances of Nature coming to the rescue of Man, and even the Gods, sometimes. For instance, in the Ramayana, Ram's brother Lakshman is saved by the Sanjeevani herb that Hanuman brings from the foothills of the Himalayas. Although the story is a metaphor for *Dasyabhakti* and filial love, it showcases the importance of medicinal herbs in curing mankind of its many maladies. India's ancient herbal medicine Ayurveda makes use of the power of plants to cure disease. Sushruta, who penned the medical treatise *Sushruta Samhita*, says that God himself incarnated as the king of Varanasi, Dhanvantari, to teach medicine to physicians.

These instances of valor and volumes on medical lore are part of the Indian ethos, but today, the venomous skirmish between Man and Nature has triggered the human survival at risk at the backdrop of ecological imbalance and obliteration of biodiversity. In this context the writings of Vandana Shiva make a very compelling argument for a more sensible approach in exploiting Mother Earth's bountiful offerings. The environmental activist and food-sovereignty advocate has authored more than 20 books of which *Earth Democracy* (2005), *Soil not Oil* (2007) and *Making Peace with the Earth* (2012) make compelling reading because of their 'pro-environment' and 'women empowerment' arguments.

This paper attempts in assessing Vandana Shiva's role as an ecofeminist in establishing her credentials as a green rights activist and a committed writer who fights for woman's role in protecting the ecological balance as well as taking up the cudgels against the destruction of biodiversity.

Regression is a tool to Measure the Working Capital Performance - Impact on Profitability of Banks: A Case Study of SBI & HDFC Bank Ltd.

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Article Info

Volume 83

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Abstract

Profitability is the financial measure of firm's ability to earn profit. It can be measured through Working Capital Performance, such as current assets and current liabilities. It is important for banks to have efficient working capital practices as a portion of funds is parked to meet short term financial obligations to stakeholders of banks and ability to continue its operations. The idea of this paper is to examine the working capital impact over the profitability used a statistical tool with regression analysis to measure the financial performance of the SBI and HDFC top ranked two banks in India. These banks, shows a tremendous growth over the year. Data for these two banks were collected for five years from 2013-14 to 2018-19 from secondary sources and analyzed. It was found out that the relationship between working capital performance with profitability ratios consisting of gross profit ratio, net profit ratio, operating profit ratio, return on equity, return on capital employed and earnings per share of the two select banks. It also studies the relationship between these variables using regressions, z- test, Correlation and used t- test for hypothesis testing. Findings of this study all profitability components are uniformly influenced the working capital performance of SBI and HDFC.

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Keywords; Working Capital management, Profitability, Regression Analysis, Return on Equity, Return on Capital Employed, EPS, etc.

I. INTRODUCTION

Every business needs funds for two purposes for its establishment and to carry out its day to day operations. Working capital refers to that part of the firm's capital which is required for financing short term or current assets such as cash, bank, marketable securities, bills, receivables, debtors and inventories. Working capital is the required amount of funds needed to cover the cost of operating the enterprise. Working capital refers to the funds invested in current assets i.e. investment in stocks, sundry debtors, cash and other current assets. Current assets are essential to use fixed assets profitably. For example a machine cannot be used without raw material. Thus it is obvious

that certain amount of funds is always tied up in raw materials and work in progress, finished goods. However, the business also enjoys credit facilities from its suppliers who may supply raw materials on credit and the firm may not pay immediately all expenses. Therefore, certain amount of funds is automatically available to finance the current assets requirements. However, the requirements for current assets are usually greater than the amount of funds payable through current liabilities. In other words, current assets are to be kept at a higher level than the current liabilities. The object of Working capital management is to deal with the current assets and current liabilities of the firm in such a way that a satisfactory level of Working capital is maintained. Working capital is the difference between the

Regression is a tool to Measure the Working Capital Performance - Impact on Profitability of Banks: A Case Study of SBI & HDFC Bank Ltd.

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NANOTECHNOLOGY A REVIVAL APPROACH FOR SCIENTIFIC DEVELOPMENT – MATERIAL SCIENCE

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Nanotechnology is the science of engineering that deals with particle which are less than 100 nm in size. It

is the study of manipulating matter on molecular and atomic scale. In recent years, nanotechnology showed its potential in

the field of biomedical, electronics, robotics. In civil engineering and construction, the nanotechnology is applied in (i)

concrete for reducing segregation in self compacted concrete, (ii) the use of copper nano-particles in low carbon HPS is

remarkable, (iii) the use of nano sensors in construction phase to know the early age properties of concrete is very useful,

and (iv) its use in water purification system by replacing the use of granulated particles of carbon in filtration with purifiers

like Nano Ceram-Pac (NCP). The present paper reviews the state of the art on the use of nanotechnology in the field of

civil engineering and construction and also discusses its future prospect. Also, special emphasis is placed on the future

application of nanotechnology in the field of geotechnical engineering.

Abstract:

Nanotechnology is the science of engineering that deals with particle which are less than 100 nm in size. It is the study of manipulating matter on molecular and atomic scale. In recent years, nanotechnology showed its potential in the field of biomedical, electronics, robotics.

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RESEARCH ARTICLE

Condensation Methods for the Determination of Methoxamine Hydrochloride in Pure and Pharmaceutical Formulations

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ABSTRACT:

Two visible spectrophotometric methods were developed A and B for the determination of Darunavir in pure and pharmaceutical formulations. The methods are based on condensation reaction with PDAB (Method-A) and ONB (Method-B) in presence of acidic medium with the primary amine group in DNV. The coloured products exhibit absorption λ_{max} at 639 nm and 452nm for methods A and B respectively. Regression analysis of Beer-Lambert plots showed good correlation in the concentration ranges 10-60 μ g/ml, 50-300 μ g/ml, correlation coefficients are 0.9983, 0.9989; Sandell's sensitivities are 9.9833×10^{-3} , 3.0456×10^{-2} (1 mole cm^{-1}); and molar absorptivity values are 5.4857×10^4 , 1.7981×10^4 (μ g cm^{-2}) for methods-A and B respectively. The proposed methods are applied to commercial available formulations and the results are statistically compared with those obtained by the UV reference method and validated by recovery studies. The results are found satisfactory and reproducible. These methods are applied successfully for the estimation of the DNV in the presence of other ingredients that are usually present in formulations. These methods offer the advantages of rapidity, simplicity and sensitivity and low cost without the need for expensive instrumentation and reagents.

KEYWORDS: Condensation, PDAB, ONB, Regression Analysis.

INTRODUCTION:

Methoxamine hydrochloride (MHC), chemically 2-amino-1-(2,5-dimethoxyphenyl) propan-1-ol (Fig.1), is relatively selective α_1 -adrenoreceptor agonist producing an increasing in peripheral vascular resistance^{1,2}. MHC is a potent sympathomimetic that increases both systolic and diastolic blood pressure. It is used in the treatment of some hypotensive patients, particularly those with paroxysmal a trial tachycardia (arrhythmia) and acts almost exclusively on alpha-adrenergic receptors.

Literature survey reveals few analytical methods were reported for the determination of MHC in biological fluids by HPLC with fluorescence detection^{3,4}. Spectrophotometric method analysis^{5,6}. High performance liquid chromatography using pre-column derivatization of MHC with other cardiovascular drugs⁷. The analytical useful functional groups in MHC have not been fully exploited for designing suitable visible spectrophotometric methods and so still offer a scope to develop more visible spectrophotometric methods with better sensitivity, precision and accuracy.

No analytical methods that have been reported so far for the estimation of MHC by HPLC method. Hence, the author has made an attempt to develop a new, simple, economical, selective, accurate and precise stability indicating reverse phase high-performance liquid chromatographic method⁸⁻¹² with good sensitivity for assay of MHC in pharmaceutical dosage form. Indeed the established method was validated with respect to

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A Negotiation of Home, Reminiscence, and Identity in M G Vassanji's *The Assassin's Song*: A Humanistic Geographical Outlook

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Abstract

This paper attempts to reconnoiter the notions of home, reminiscence and identity in M G Vassanji's *The Assassin's Song* at the backdrop of humanistic geography which ascended as an antiphon to spatial sciences during the 1950s and 1960s. The term 'home' connotes the meaning associated with the basic impression encompassing location, an area, or a specific direction of human existence. Under the demesne of humanistic geography, 'home' or 'place' is also endowed with experiences and the sociocultural practices associated with daily life. The central line in *The Assassin's Song* is interwoven with different circumstances in the life of the protagonist, Karsan. His perception of life in Pirbaag as a youth and an aspirant student at Boston-Harvard, his marital companionship with Marge Thompson in Canada, his new-fangled identity as Krishna Fazl while he worked in a college in British Columbia, the tragedy of his only son's death, his coming back to India, the time he spent doing research at the Indian Institute of Advanced Study, Shimla and ultimately Karsan's taking over the assumed role of the Saheb (spiritual leader) of Pirbaag are a testimony to the interpretation that habitation defines one's understandings and thereby plays a pivotal role in changing and influencing the individual identity.

Keywords: sociocultural identity, humanistic geography, implication of place, reminiscences, new dimensions

A Negotiation of Home, Reminiscence, and Identity in M G Vassanji's *The Assassin's Song*: A Humanistic Geographical Outlook

The present-day perilous scenario in literature has countersigned a metamorphic conversion with innumerable theories of analysis, interpretation of literary texts and probing insights into the novel approaches with the existing time. The explication of a specific text is thoughtfully contrived from an extensive outlook which productively compels further latitude for research. Contrasting Vassanji's other novels which are found to be positioned chiefly in Kenya or Tanzania, *The Assassin's Song* by M G Vassanji is principally set in India. The

PROBLEMS AND PROSPECTS OF STREET VENDORS: A STUDY WITH REFERENCE TO VISAKHAPATNAM CITY

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ABSTRACT

Street vendors are an integral part of urban economies around the world, offering easy access to a wide range of goods and services in public spaces. Even though street vendors are considered as informal, they are the vital contributors to Urban economies. In this 21st century most of the people are street vendors. The Informal Economy Monitoring Study (IEMS) revealed, ways in which street vendors are strengthening their communities. Street vending plays a major role in employment creation, production, and income generation. Street vendors are facing many problems in the workplace from the public, police personnel, politicians and local rowdies. The study investigated their Problems and prospects within the Visakhapatnam beach region. The government of India (Ministry of Housing and Urban Poverty Alleviation) passed the street vendors Bill 2013 in order to improve their standard of living. Street trade also adds vibrancy to urban life and in many places is considered a cornerstone of historical and cultural heritage. In this study, the researchers evaluated the problems faced by street vendors in their workplace and the prospects needed to get facilitated. The researchers selected 110 respondents from R.K.Beach to Bheemili Beach, located in Visakhapatnam Corporation by adopting convenient sampling and the study is descriptive in nature. Interview method was used to collect data. This study aims to find out the problems faced by the street vendors and prospects needed, in order to protect their rights and do their street vending business. The paper provides spatial solutions for policy measures for regulating these informal entrepreneurs.

Keywords: Informal entrepreneurs, Street Vendor Problems, Informal Economy Monitoring Study, Urban economies, cultural heritage.

1. Introduction

An entrepreneur has to function within limits of freedom and has to choose from limited opportunity. Key element of successful entrepreneurship is to make prudent choices from those available, and act upon the choices incrementally (not at one go) at various stages of development of the business, these choices are neither fully subject to environmental forces, nor to the absolute free will of the entrepreneur alone (Ray and Ramchandra, 1996).

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ABSTRACT

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Kernel System of an Automobile Service Station

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Abstract- This experimental work defines the effectiveness of chemical and physical methods in treating the wash water collected from automobile service stations. Wash water was collected from a service stations in the city of Visakhapatnam, and the effluent was characterized for different parameters such as pH, total hardness, turbidity, total dissolved solids, oil and grease, detergents. For chemical treatment, alum was used as coagulant. Sugarcane bagasse and wood chips were used as filter media for physical treatment. Alum at different concentrations was agitated with the effluent for a prescribed contact period (30 min) which led to the formation of flocs. The sifted tests were tried and the rate decrease in oil and grease was connected with the profundity and sort of filter media. The physical and concoction strategies for treatment uncovered that normal materials are compelling in evacuating real parameter like oil and oil and cleansers from the vehicle emanating and subsequently give a practical arrangement in view of its eco-amicability. From the exploratory investigations it is seen that in physical treatment. As adsorption procedure, the adsorption limit of any material is reliant on its porosity, specific surface zone and thickness of filter bed.

Keywords: Kernel system, Automobile service stations, Physical treatment, Adsorption technique, Specific surface area.

1. INTRODUCTION

On location wastewater treatment is an imminent heading towards the decrease of contamination load to the civil or consolidated urban wastewater. One of such territories is car benefit station and vehicle washing. A portion of the current petroleum siphons, fuel benefit stations are encouraged with vehicle washing the large Car washing pools of garage and service stations are insisted upon by the respective PCB's to provide necessary treatment for the effluents. Proficient vehicle wash frameworks make wastewater that can greatly affect the earth, if not appropriately oversaw and released. Contaminants in wash wastewater incorporate oil and grease, detergents, phosphates, hydrofluoric acid, ammonium bi-fluoride items (ADF) etc. Oil and grease and detergents, including biodegradable cleansers can be harmful to aquatic animals. Then again, Phosphates, which are plant supplements, can cause inordinate development of disturbance plants like microscopic plants.


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Speed Control of BLDC Machine Using Different Control Strategies

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Abstract - Brushless DC motors Controlled through electronic commutation whose motor position is sensed using Hall Sensors. In this paper, initially the responses of BLDC motor are observed by using PI and SMC Controllers. The Sensitivity of PI towards the constraint variations and unexpected torque disturbances, PI Controller replaced with Sliding Mode Controller (SMC). SMC is an effective synthesis technique for Non-Linear uncertain systems with fast and robust transient responses. This paper deals with a new control technique based on PWM control of brushless DC motor (BLDC). PWM technique is broadly used in motors which are controlled using power converters. Also, it controls analog systems with digital processing outputs.

Key Words: BLDC: Sliding Mode Control (SMC): PWM.

1. INTRODUCTION:

A brushless DC motor (BLDC) is a PMSM motor with trapezoidal back emf and electronically commutated system [1]. The permanent magnet synchronous motors (PMSM) have the advantage of high efficiency, high torque density, less volume and less maintenance which are satisfied with EV accurate necessities [2-3]. BLDC motors have less noise and large durability. The main advantage of this motor is that it can be controlled by using feedback mechanisms for faster speed responses and less ripples in torque. In conventional PI controller, the motor performance may result disturbances in torque due to variations in parameters and its sensitivity towards the uncertainty nature of

the system. In order to conquer this disadvantage, sliding mode control (SMC) is adopted. Generally, the practical BLDC motor is nonlinear and there are number of disturbances, to overcome these disadvantages and to improve the performance of motor SMC is adopted. Even though SMC is vigorous to changes that occur in motor, the increment of gains which is used for controlling purpose causes chattering effect which is not desirable and causes ripples in the responses and also results high frequency switching in converters. To overcome these disadvantages PWM techniques with SMC controllers is implemented in this paper. PWM technique encodes the analog value to digital values by controlling duty cycle the method of implementing SMC controller: 1. Initially observing the convergence and setting of the control law. 2. Later establish the sliding mode at the initial time to give the robust behaviour of the implemented control law all over the system response. 3. The making of the path enabling the convergence in finite time [5], [6]. In this work overview of control of BLDC motor using PI controller and sliding mode control is observed. Later the control of motor with SMC PWM technique is observed. The results obtained in simulation and a conclusion where we emphasize the interest of this method of control.

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